

12-Month Backend Engineering Roadmap

PHASE 0 (Weeks 1–4): Backend Fundamentals

- Transactions & isolation levels
- Indexing & execution plan analysis
- REST API best practices
- Logging & exception handling
- JUnit & integration testing

PHASE 1 (Weeks 5–8): Docker Mastery

- Images, layers, multi-stage builds
- Networking & volumes
- Docker Compose (App + DB)
- Rebuild from scratch without tutorial

PHASE 2 (Weeks 9–12): CI/CD

- GitHub Actions or GitLab CI
- Build + test + docker pipeline
- Push to container registry
- Branch protection workflow

PHASE 3 (Weeks 13–28): AWS Cloud

- IAM, EC2, Security Groups
- RDS deployment
- ECS + ECR
- S3 + CloudWatch
- Load balancing basics

PHASE 4 (Weeks 29–36): Kafka Deep Dive

- Partitions & consumer groups
- Rebalancing & ordering guarantees

- Retry & DLQ strategies
- Build mini event-driven system

PHASE 5 (Weeks 37–48): Advanced Engineering

- Redis caching
- Distributed locking
- Observability & metrics
- System design fundamentals

WEEKLY RETENTION RULE:

- Rebuild 1 component every Sunday
- Redeploy project monthly
- Review past phases quarterly